Machine Learning

Mathematics and Computing

Tutorial 3

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Exercise 1: Knowledge Flow

After executing the KnoledgeFlow, what information is shown on each of them? What is percentage of correctly classified instances?

The first TextViewer (connected to J48) displays the textual representation of the J48 decision tree models. It contains 10 entries, one for each fold of the cross-validation, showing the tree structure with its decision rules, number of leaves and tree size. The second TextViewer (connected to the Classifier PerformanceEvaluator) displays the evaluation results of the J48 classifier across all 10 folds of the cross-validation. This includes a summary with correctly and incorrectly classified instances, Kappa statistic, error metrics, detailed accuracy by class and lastly the confusion matrix. The percentage of correctly classified instances is 86.2105 %.

What is the utility of creating knowledge flows with this Weka interface?

This interface, unlike the Explorer module, allows us to visually design, execute, and reuse data mining workflows by dragging and connecting components on a screen. It also offers greater flexibility than the Explorer, and provides a more visual and helpfull representation of the entire pipeline.

Exercise 2: Experimenter

Is there a data set that seems more appropriate?

Yes, the dataset lunar lander filter1 appears to be the most appropriate. It achieves the highest overall accuracy of 90,96 % with the IBk algorithm ($k = 1$). Compared to the original dataset and lunar lander filter2, this version provides the best predictive performance across the most competitive models.

Which algorithm do you think is the best?

The IBk (K-Nearest Neighbors) algorithm with $k = 1$ is the best performer. It consistently achieves the highest accuracy across all three datasets (90,56 %, 90,96 %, and 84,81 %). Furthermore, it is marked with a 'v' in all rows, indicating that it is statistically superior to the base J48 model in every instance.

Are the results of the best algorithm much better than the others?

The results of IBk ($k = 1$) are significantly better than the baseline and simple models, but only marginally better than IBk with $k = 3$. While it outperforms ZeroR (75,04 %) and NaiveBayes ($\approx$ 75 %) by roughly 15 %, the difference between $k = 1$ and $k = 3$ is less than 1,5 %. However, since it is the only one to consistently maintain the highest percentage across all filters, its superiority is statistically confirmed.

What do you think the Weka Experimenter is suitable for?

We think the Weka Experimenter is really useful because it lets you run a lot of different tests at the same time without having to set up each one manually. Instead of testing one algorithm at a time, you can just add a bunch of them like J48, IBk, or NaiveBayes and run them across several datasets all at once to see which one works best. It is also great for checking if the results are actually better or just lucky by using a statistical which marks the improvements with a 'v' or a '*'. Finally, it is very handy because it saves everything into one file.

Conclusions

In this tutorial, we have explored two advanced Weka interfaces that streamline the process of machine learning experimentation. Through the **KnowledgeFlow** exercise, we demonstrated how to graphically design a data processing pipeline, which provides a clear and reusable visualization of the entire workflow from data loading to evaluation. This approach is particularly efficient for understanding the interaction between different components in a cross-validation process.

Furthermore, the **Experimenter** module proved to be an essential tool for statistical comparison. It allowed us to evaluate multiple algorithms (specifically J48, IBk, PART, ZeroR, and NaiveBayes) across different datasets simultaneously. We determined that IBk with $k = 1$ was the most effective model, showing statistically significant improvements over the baseline. Overall, these interfaces enhance our ability to conduct rigorous, large-scale experiments and ensure that the selected models and data filters are truly the most appropriate for the task.